Dragomir Božoki

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Summary

M.Sc. student in Computer Science (Vision and Graphics) at Sorbonne Université, with industry experience building and deploying real-time multimodal AI systems for humanoid robots and production-grade LLM applications. Research background in speech-driven facial animation, visual speech recognition and multilingual NLP, bridging applied machine learning with multimodal learning. Available for a 6-month internship from mid-February 2027 (convention de stage).

Research interests: computer vision, audio-visual learning, speech and language technologies, efficient deep learning and on-device inference.

Experience

AI Engineer, *Comtrade System Integration, Belgrade, Serbia* Oct 2025 – Sep 2026

- Cut end-to-end voice response latency on humanoid robots to 2–2.5 s, including retrieval and synthesis, by building a streaming ASR, LLM and neural TTS pipeline (LiveKit) with barge-in and addressee detection (write-up)
- Owned the on-robot conversational and behavioural stack on NVIDIA Jetson Orin: ROS 2 bridge to Python AI services, audio bridge with acoustic echo fixes, and dialogue-driven gestures coupled with camera-based perception
- Developed agentic AI workflows combining LLM-based reasoning, tool orchestration and context-aware information retrieval for cloud and on-premises deployments, with self-hosted models under data-residency constraints
- Shipped the surrounding stack end to end: Python/FastAPI services, C#/.NET middleware and React operator dashboards, behind feature flags with regression tests and latency budgets

Machine Learning Intern, *Vinaver Medical, Novi Sad, Serbia* Oct 2024 – Feb 2025

- Built a custom CNN for 3D segmentation of CT images, complemented by an analytical method I derived and implemented alongside the deep learning approach; presented the work at a startup seminar in Sofia

Education

M.Sc. Computer Science, Vision and Graphics (IMA), *Sorbonne Université, Paris* Sep 2026 – Sep 2027

Taught in English, co-accredited with Télécom Paris: image analysis, computer vision, 3D graphics, deep learning.

M.Sc. Electrical Engineering, Signal Processing, *University of Novi Sad, Serbia* Oct 2024 – Aug 2026

GPA 9.1/10. Thesis on multilingual intent understanding. Erasmus+ exchange, University of the Peloponnese, Greece.

B.Sc. Biomedical Engineering, *University of Novi Sad, Serbia (GPA 8.1/10)* Oct 2020 – Jul 2024

Selected Research & Projects

Audio-Visual Avatar Animation via Blendshape Prediction from Speech and Text in Serbian write-up

D. Božoki, I. Marković, S. Suzić, N. Šerović, A. Dušanić, T. Nosek | In preparation for TELFOR 2026 (IEEE)

Develops a speech-driven avatar animation system predicting 52 ARKit blendshape coefficients from Serbian speech and text. Combines causal TCNs for real-time inference, BiGRUs for offline prediction, attention-based text fusion and viseme supervision. 2nd place, AI-SPEAK competition.

Multilingual Hybrid NLP for Semantic Intent Understanding M.Sc. thesis, 2026 | code | write-up

Built an XLM-RoBERTa intent classifier achieving 97.1% accuracy and 0.967 macro-F1, with FAISS-based semantic retrieval as a fallback, integrated into the open-source Kalliope voice assistant.

Visual Speech Recognition from Silent Video B.Sc. thesis, 2024 | code | write-up

Developed sentence-level lip reading on GRID (22 speakers) using a 3D-CNN, BiGRU and CTC loss, with a custom dlib/OpenCV landmark pipeline. Achieved 7.3% character error rate (CER) and 15.7% word error rate (WER).

Chest X-ray Generation with GANs University project, 2026 | code | write-up

Implemented a convolutional GAN with custom generator, discriminator and adversarial training loop to generate 512×512 chest X-rays from 256-dimensional latent vectors, using exponential moving average (EMA) fine-tuning.

Skills

Deep learning: PyTorch, TensorFlow, scikit-learn, pandas, NumPy

Speech and NLP: streaming ASR, TTS, XLM-RoBERTa, FAISS, hybrid retrieval, reranking, vLLM, LangChain

Systems: ROS 2, NVIDIA Jetson Orin, LiveKit, Docker, FastAPI, Azure, on-premises GPU inference, CI/CD, Git

Programming: Python, C, C#, SQL, MATLAB, JavaScript, React

Awards and Languages

- 2nd Place, AI-SPEAK Competition, FTN Novi Sad (2026) --- blendshape prediction from speech
- BOLD Fellow, Network 20/20 and University at Buffalo (2023) --- entrepreneurship programme; pitched at Goldman Sachs HQ, NYC

Languages: Serbian (native), English (fluent), French (beginner), German (A2), Hungarian (basic)